

LESSON 6 : PERCENTAGES II (C)

Checkpoint 1

Simple Interest :

$$I = P \times R\% \times T$$

$$A = I + P$$

$$A = (P \times R\% \times T) + P \text{ or } P(1 + R\% \times T)$$

P = Principal

R = Rate

T = Time (by year)

I = Interest

A = Amount

1. Find the simplest interest of the following.

- | | | |
|--------------------------|--------------------------|-------------------|
| (a) Principal: \$2 000; | Interest rate 4% p.a.; | Period: 3 years |
| (b) Principal: \$300; | Interest rate 3.5% p.a.; | Period: 1.5 years |
| (c) Principal: \$60 000; | Interest rate 2% p.a.; | Period: 8 months |

2. Johnny deposits \$ 2 000 in Bank *A* at 10 % p.a., and \$ 3 500 in Bank *B* at 7 % p.a.

- Find the simple interest he will receive in Bank *A* each year.
- Find the simple interest he will receive in Bank *B* each year.
- Find the total interest he will receive each year.

3. Raymond deposits \$24 000 in a bank at a simple interest rate of 3% p.a. If he increases the deposit period from 2 years to 5 years, find the increase in the interest earned.

- ◆ 4. Shirley deposits \$50 000 at a simple interest in a bank for 4 years. If the interest rate is increased from 4% to 9%, find the increase in the interest earned.
5. If \$8 000 is deposited in a bank at a simple interest rate of 2% p.a., how long will it take to receive an interest of \$960?
- ◆ 6. Sam borrows \$20 000 from a bank at a simple interest rate and he has to repay an amount of \$21 200 after 2 years. What is the annual interest rate?
7. If \$32 000 is deposited in a bank at a simple interest rate of 3% p.a., how long will it take to receive an amount of \$34 880?
8. Alex borrows \$ P from a bank at a simple interest rate 5% and he has to repay an amount of \$ 19550 after 3 years. Find the value of P .

9. A sum of \$12 000 is deposited at a simple interest rate of 3% p.a. For each of the following deposit periods, find

- (i) the simple interest,
- (ii) the amount.

- (a) 2.5 years
- (b) 18 months

Hints: 18 months = ? year

10. Mrs. Yeung deposits some money in a bank at an interest rate of 3% p.a.. She gets \$600 simple interest for the first quarter of the year.
- (a) How much does Mrs. Yeung deposit at the beginning?
 - (b) If the interest rate is increased by 3 % p.a., find the simple interest that she will get for the remaining 3 quarters of the year.

11. If the interest rate is 12% p.a. and the simple interest repaid after 15 months is \$1 500, find the principal borrowed.

- ◆ 12. Maggie deposits \$15 000 in Bank *A* at a simple interest rate of 6% p.a. for 1.5 years, then she deposits the amount in Bank *B* at a simple interest rate of 3% p.a. for 2 years. Find
- (a) the amount she obtained from Bank *A*,
 - (b) the amount she obtained from Bank *B*,
 - (c) the total interest earned from both banks.

Checkpoint 2

Compound Interest :

$$A = I + P$$

$$A = P\left(1 + \frac{R\%}{n}\right)^t$$

$$I = P\left(1 + \frac{R\%}{n}\right)^t - P$$

P = Principal

R = Rate

n = number of Times (by year)

t = number of times during the period

I = Interest

A = Amount

13. \$10 000 is invested at 10% p.a. for 3 years. If the interest is compounded yearly, find

- (a) the amount,
- (b) the compound interest after 3 years.

◆ 14. Tracy deposits \$30 000 in a bank at 3% p.a. compounded yearly, find the compound interest after 4 years.

15. Mr Tam invests \$8 000 for 3 years at 12% p.a. Find the amount he will obtain if the interest is compounded

- (a) half-yearly,
- (b) quarterly,
- (c) monthly.

16. Kate deposits \$ P in a bank at 6% p.a. and the interest is compounded half-yearly. If she obtains an amount of \$23 880 after 3 years, find
- (a) the value of P ,
 - (b) the compound interest earned.
17. Steve deposits a sum of money in a bank at 8% p.a. and the interest is compounded yearly. If he obtains an amount of \$99 144 after 2 years,
- (a) find the principal,
 - (b) If the annual interest rate is unchanged and the interest is compounded monthly, find compound interest he will obtain after 9 months.
- ◆ 18. Alice deposits \$30 000 individually in two banks A and B . Bank A offers an interest rate of 6% p.a. compounded yearly, while bank B offers the same annual interest rate compounded quarterly. Assume that both deposits are for 3 years. Find
- (a) the amount she obtained from bank A ,
 - (b) the amount she obtained from bank B ,
 - (c) the difference in the amounts she obtained from the two banks.

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19. A principal of \$50 000 is deposited for 2 years at an interest rate of 6% p.a.. Find the amount if it is compounded:
- (a) half-yearly;
 - (b) quarterly;
 - (c) monthly;
 - (d) daily (assume there are 365 days in a year).
- (Correct your answers to the nearest dollar.)
20. Daniel deposits \$48 000 in a bank for 2 years and the interest is compounded monthly. If the interest rate is increased from 4% to 5%, find the increase in the amount obtained.
- ◆ 21. Chris deposits \$5 000 in bank *A* at 10% p.a. compounded half-yearly for 2 years, then he deposits the amount in bank *B* at a simple interest rate of 5% p.a. for $1\frac{1}{2}$ years. Find
- (a) the amount he obtained from bank *A*,
 - (b) the total interest earned from two banks.
22. \$3 000 is deposited for 5 years at an interest rate of 10% p.a., find the different between the simple interest and the compound interest compounded yearly.

Checkpoint 3

A man borrows \$ 5 000 from a bank at 12 % p.a., compounded monthly. He repays \$ 500 at the end of each month. Find how much he still owes the bank after the second repayment, correct to 1 decimal place.

May		June		July	
At the beginning e.g. 1/5/2020	\$5000 <i>is borrowed from the bank</i>	Then he repays \$500 on 1/6/2020	After repayment, \$5050 – 500 = 4550 <i>borrowed from the bank</i>	Then he repays \$500 on 1/6/2020	After repayment, \$4595.5 – 500 = 4095.5 <i>borrowed from the bank</i>
31/5/2020 And before repay the money	Total money $\text{\$5000} \left(1 + \frac{12\%}{12} \right)^{(12)(1/12)}$ = \$5050 <i>borrowed from the bank</i>	30/6/2020 And before repay the money	Total money $\text{\$4550} \left(1 + \frac{12\%}{12} \right)^{(12)(1/12)}$ = \$4595.5 <i>borrowed from the bank</i>		

23. A man borrows \$20 000 from a bank at 10% p.a. and repays \$5 000 at the end of each year. Find the amount he still owes after the fourth payment.

24. A man borrows \$5 000 from a bank at 12% p.a., compounded monthly. He repays \$500 at the end of each month. Find how much he still owes the bank after the second repayment, correct to 1 decimal place.

25. Jason borrowed \$35 000 from a bank at an interest rate of 18% p.a. compounded monthly, and he had to repay \$10 000 at the end of each month.
- How much, correct to the nearest dollar, did he owe the bank at the beginning of the third month?
 - How many repayments were needed to settle the loan?

Checkpoint 4

Miss Kwan's total annual income for the year 2019/20 is \$ 300 000. If her salaries tax allowance is \$ 120 000, how much salaries tax should she pay

Step 1 Covert total annual income into Net chargeable income

$$\begin{aligned}\text{Net chargeable income} &= 300\,000 - 120\,000 \\ &= \$ 180\,000\end{aligned}$$

Step 2 How many \$ 50 000 can be divided from
Net chargeable income?

ANS: 3 and the remainder is \$ 30 000

Step 3 Calculation

$\therefore$ Miss Kwan's salaries tax payable

$$\begin{aligned}&= 50\,000 \times 2\% + 50\,000 \times 6\% + 50\,000 \times 10\% + 30\,000 \times 14\% \\ &= \$ 13\,200\end{aligned}$$

Net chargeable income	Tax rate	Salary tax
On the first \$ 50 000	2 %	\$ 1 000
On the next \$ 50 000	6 %	\$ 3 000
On the next \$ 50 000	10 %	\$ 5 000
On the next \$ 50 000	14 %	\$ 7 000
Remainder	17 %	

26. The net chargeable income of Thomas is \$155 000. How much salaries tax should he pay?

27. If the basic allowance for a single person is \$108 000, find the amount of salaries tax for each of the following monthly income.

(a) \$10 000

(b) \$63 000

28. Miss Lo's salaries tax payable charged at progressive rates is \$240 for this year. If she has an allowance of \$216 000, find her annual income.